DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	**ITCP412**
COURSE TITLE	**:**	**Capstone Project 2**
COURSE TYPE	**:**	Lecture / Research
COURSE CREDIT	**:**	3 Units
LEARNING MODALITY	**:**	Traditional Blended Learning Model
PRE-REQUISITES	**:**	ITCP322 – Capstone Project 1
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Paulino H. Gatpandan, DIT phgatpandan@dlsud.edu.ph
CONSULTATION HOURS	**:**	0800 – 1000 Monday (COS100C), Thursday (Online) 1130 – 1200 Monday (COS100C), Thursday (Online) 1300 – 1430 Monday (COS100C), Thursday (Online) 1600 – 1700 Monday (COS100C) 1100 – 1200 Wednesday (Online)

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course requires the fulfillment of the complete documentation, software, and oral defense before a set of panelists.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Understand the latest concepts in systems analysis and design, software engineering and information technology.
CLO2. Analyze original research data and present the data in a form appropriate for the design.
CLO3. Collect original data following ethical protocols.
CLO4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.

CLO5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Understand the latest concepts in systems analysis and design, software engineering and information technology.	TLO1. Analyze complex problems, identify, and define the computing requirements appropriate to its solution. TLO2. Identify and analyze user needs and take them into account in the selection, creation, and evaluation and administration of computer-based systems. TLO3. Study and apply various high level and detailed design and implementation principles for building a software system.
CLO 2. Analyze original research data and present the data in a form appropriate for the design.	TLO4. Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.
CLO 3. Collect original data following ethical protocols.	TLO5. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO 4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.	TLO6. Function effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO 5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.	TLO8. Apply knowledge through the use of current techniques, skills, tools, and practices necessary for the IT profession. TLO9. Develop a formal Presentation of the research project and engage/answer questions in an oral presentation
CLO 6. Demonstrate 21st century skills in all learning activities.	TLO10. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Module 1 CLO1 CLO6 TLO1 TLO2 TLO3 TLO10	Week 1 Jan 20-25	**Course Orientation** • Gospel Reading, Reflection and Prayer • Course Overview • Course Polices and Requirements • Professor Profile • Grading System • Netiquette • Group meeting	Schoolbook MS Teams Power Point	1.5	1.5	3.0
Module 2 CLO2 CLO3 CLO6 TLO4 TLO5 TLO10	Week 2 Jan 27 - Feb 1	**Chapter 4 Results and Findings** • Gospel Reading, Reflection and Prayer • Writing the results and findings • Cross Validation • Data Validation and Consultation with the Adviser	Schoolbook MS Teams Power Point	1.5	1.5	3.0
Research Work CLO2 CLO3 CLO4 CLO5 CLO6 TLO4 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 3-5 Feb 3 - 22	• Gospel Reading, Reflection and Prayer • Journal Manuscript Writing • System Development • Consultation with the Adviser	Schoolbook MS Teams		9.0	9.0
Module 3 CLO3 CLO4 TLO5 TLO6	Week 6 Feb 24 -	**Chapter 5 Summary, Conclusions and Recommendations** • Gospel Reading, Reflection and Prayer • Writing the summary, conclusions and recommendations	Schoolbook MS Teams Power Point Schoolbook	1.5	1.5	3.0

TLO7	Mar 1	**Enabling Assessment:** (Not Graded) Submission of Progress Report and Ethics Certificate				
Research Work CLO2 CLO3 CLO4 CLO5 CLO6 TLO4 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 7-9 Mar 3 - 22	• Gospel Reading, Reflection and Prayer • Journal Manuscript Writing • System Development • Consultation with the Adviser **Enabling Assessment:** (Not Graded) Submission of Progress Report	Schoolbook MS Teams		9.0	9.0
			Total	**4.5**	**22.5**	**27**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Collect original data following ethical protocols.	TLO5. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.	TLO6. Function effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7 Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.	TLO8. Apply knowledge through the use of current techniques, skills, tools, and practices necessary for the IT profession. TLO9. Develop a formal Presentation of the research project and engage/answer questions in an oral presentation.
CLO6. Demonstrate 21st century skills in all learning activities.	TLO10. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Module 4 CLO4 TLO6 TLO7	Week 10 Mar 24-29	**Publishing Paper – Article Writing** • Gospel Reading, Reflection and Prayer • IMRAD Guide • Article Writing	Schoolbook MS Teams	1.5	1.5	3.0
Research Work CLO3 CLO4 CLO5 CLO6 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 11–12 Mar 31 - Apr 12	• Gospel Reading, Reflection and Prayer • Consultation with Advisers • Journal Manuscript Writing • System Development	Schoolbook		6.0	6.0
	Week 13 Apr 14-19	Independent Learning Days / Holy Week	Schoolbook		3.0	3.0
Research Work CLO3 CLO4 CLO5 CLO6 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 14-15 Apr 21 - May 3	• Gospel Reading, Reflection and Prayer • Consultation with Advisers • Journal Manuscript Writing • System Development • Submission of Progress Report and Endorsement Form of Adviser • Journal review of published papers of faculty members indicated in the references	Schoolbook		6.0	6.0

Final Defense CLO5 CLO6 TLO8 TLO9 TLO10	Week 16-18 May 5 - 24	**Summative Assessment**: Oral Presentation (Final)		9.0		9.0
			Total	**10.5**	**16.5**	**27**

GRADING SYSTEM:

Finals
Enabling Assessments (Adviser's Grade) - 20%
Summative Assessments (Panel's Grade) - 80%
100%

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Call number or e-provider	**References**
	Undergraduate Research Handbook: Policies and Guidelines
QA 402 .D423a 2019	Dennis A, Wixom B, & Roth R. 2019. Systems analysis and design. 7th edition. Hoboken, NJ : Wiley
QA 76.9.D26 .C816 2019	Coronel C, Morris S. 2019. Database systems: design, implementation and management. 13th edition. (Australia, United States): Cengage Learning
ebooks	Ariola M. 2006. Principles and methods of research. Rex Bookstore Inc
Journals	Z. Gao, M. P. Gatpandan and P. H. Gatpandan, "Classification Decision Tree Algorithm in Predicting Students' Course Preference," *2021 2nd International Symposium on Computer Engineering and Intelligent Communications (ISCEIC)*, Nanjing, China, 2021, pp. 93-97, doi: 10.1109/ISCEIC53685.2021.00026. J. Xu, P. H. Gatpandan, M. P. Gatpandan and Z. Gao, "Freight and Passenger Volume Prediction using Multiple Regression," *2021 International Conference on Machine Learning and Intelligent Systems Engineering (MLISE)*, Chongqing, China, 2021, pp. 77-83, doi: 10.1109/MLISE54096.2021.00022. X. Gao, M. F. Rosas, Y. Zhang, M. P. Gatpandan, P. H. Gatpandan and D. Dasig, "The Research and Analysis of Student Health Status on the Basis of Physic Fitness Test' Data," *2021 International Conference on Public Health and Data Science (ICPHDS)*, Chengdu, China, 2021, pp. 110-115, doi: 10.1109/ICPHDS53608.2021.00031.

	D. J. R. Calantoc, D. D. Dasig, E. E. Claricia, M. A. B. Taduyo, P. H. Gatpandan and R. V. F. Guarin, "Clustering and Predicting Smartphones Features using Gaussian Mixture Model Algorithm," *2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, Manila, Philippines, 2021, pp. 1-6, doi: 10.1109/HNICEM54116.2021.9731964.

Prepared: **Information Technology Department**
2nd Semester 2024-2025

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Chair, ITD

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cf: Office of the Provost, AEA, CICS Curriculum Development Committee, file